COMPUTER SCIENCE

Maximum Marks=100 **Time: 3 Hrs**

Theory =70 Marks **Practical =30 Marks**

1.	Computer Fundamentals	10 marks
2.	Software Concepts	10 marks
3.	Number System	10 marks
4.	Programming methodology	10 marks
5.	Introduction to Python	10 marks
6.	Data types and operators	10 marks
7.	Strings in Python	10 marks

UNIT 1: COMPUTER FUNDAMENTALS

- History of Computers
- Generations of Computers
- Functions of a Computer
- Block diagram of a Computer system
- Brief description of following functional components of a Computer system:

- Input devices: Keyboard, Mouse, Scanner, barcode reader
- Output devices: Monitor, Printer
- CPU: ALU and CU
- Memory unit
- Primary memory: Cache memory, RAM, ROM
- Secondary memory: Hard disk drive, CD, DVD, Pen drive
- Units of Memory: Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte Peta Byte
- Concept of PROM, EPROM, EEPROM

UNIT 2: SOFTWARE CONCEPTS

- Definition of Software
- Types of software (System Software, Application Software, Utility Software)
- Need for Operating System
- Functions of Operating System (Processor management, Memory management, File management, Device management)
- Concept of computer languages: Machine language, Assembly language, High level language.
- Language Processors: Assembler, Compiler and Interpreter

UNIT 3: NUMBER SYSTEM

- Number Systems: Decimal, Binary, Octal, Hexadecimal
- Conversion from Decimal number system to Binary, Octal and Hexadecimal number system (Whole numbers only)
- Conversion from Binary, Octal and Hexadecimal number system to Decimal number system (Whole numbers only)
- Conversion from Binary number system to Octal, Hexadecimal number system using shortcut method (whole numbers only)
- Conversion from Octal, Hexadecimal number system to Binary number system using shortcut method (whole numbers only)

UNIT 4: PROGRAMMING METHODOLOGY

- Concept of a Program
- Characteristics of a good program
- Concept of Modular approach
- Program Documentation (Internal & External documentation)
- Program Maintenance
- Debugging a program
- Error and types of errors (Syntax error, Logical error, Runtime error)

UNIT 5: Introduction to Python

- Origin of python
- Unique Features of python
- Python character set
- Tokens (Keywords, Identifiers, Litrals, Operators, Punctuators)
- The print and input statement
- comments in python
- Concept of variables
- Rules for naming a variable

UNIT 6: Data Types and operators

- Built-in data types in Python
  - Text type: str
  - Numeric type: int, float
  - Sequence type: list
  - Boolean Type: bool
- Displaying the data type (use of type)
- Type Conversion (from int to float and float to int)
- Operators
  - Arithmetic operators (+, -, *, /, %, **)
  - Comparison operators (==,!= ,>,>=,<,<=)
  - Logical operators (and, or, not)
  - Assignment operators (=,+=,-=,*=,/=,%=,**=)
  - Identity operator (is, is not)
  - Membership operator (in, not in)

UNIT 7: Strings in Python

- Defining String
- Assigning string to a variable
- Multiline strings
- Use of len (), upper(), lower(),replace(),in, not in
- Slicing

- Slice from the start
- Slice to the end

➢ Use of escape characters in the string (\", \',\n,\t)

Scheme of Assessment (Theory)

The Question paper shall contain - Four sections A, B, C and D. Each section is compulsory.

1. Section A-Question 1 to 10 comprises of 10 questions of 1 mark each (MCQ's, Fill in the blanks, True/False; Assertion-Reason etc.)
2. Section B-Question 11to 19 comprises of 9 Very Short Answer (VSA)-type questions of 2 marks each.
3 Section C-Question 20 to 28 comprises of 9 Short Answer (SA)-type questions of 3 marks each.
4. Section D-Question 29 to 31 comprises of 3 Long Answer (LSA)-type questions of 5 marks each with internal choice

PATTERN	MARKS	QUESTIONS	TOTAL
Very Short Questions	01	10	10 Marks
Short Type-I Questions	02	9	18 Marks
Short Type-II Questions	03	9	27 Marks
Long Type Questions	05	3	15 Marks
TOTAL		31	70 Marks

PRACTICALS: 30 Marks (Internal=10: and External=20)

➢ **Programming in Python**

1. WAP to display "hello world".
2. . WAP to add two numbers.
3. WAP to find the length of the string.
4. WAP to convert uppercase string into lowercase and vice-versa.
5. WAP to convert temperature from Celsius to Fahrenheit
6. WAP for displaying multiline strings.
7. WAP to concatenate two strings.
8. WAP to slice a string from start to a particular position in the string.
9. WAP to replace some characters of the string with new characters.
10. WAP to find the area of a square.

➢ Practical file

Practical file must contain the entire mentioned practical.

➢ Viva voce

Viva will be asked from syllabus covered in class XI.

Distribution of 20 marks for External practical

- Programming (Logic, Syntax, documentation/ Indentation, output) (10 marks)
- Practical file (05 marks)
- Viva (05 marks)